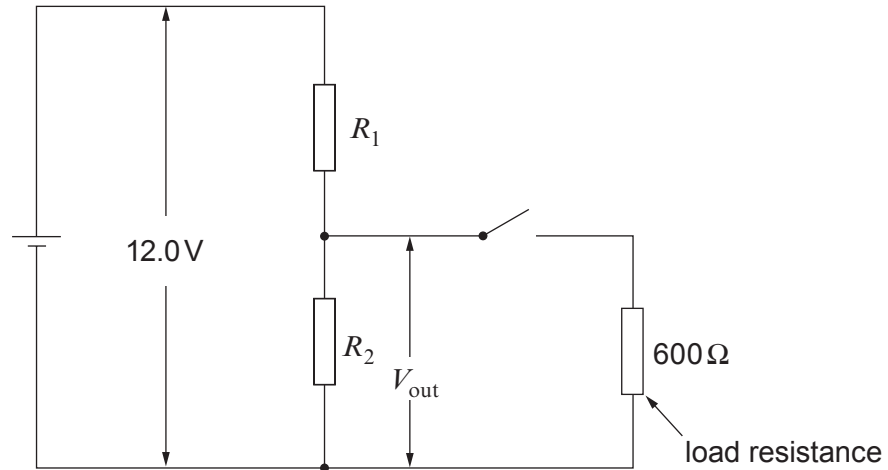


3. For the potential divider circuit shown a student is asked to select values for  $R_1$  and  $R_2$  so that the circuit meets certain requirements.



### Requirements

- A** an output pd,  $V_{out}$ , of 3.0V when unloaded (switch open),
- B** a power dissipation of less than 0.50W in  $R_1$  when unloaded (switch open),
- C** a **decrease** of less than 0.40V in  $V_{out}$  when the switch is closed.

Showing your working clearly, determine whether or not the circuit meets the requirements when  $R_1 = 450\Omega$  and  $R_2 = 150\Omega$ .

- (a) Requirement **A**: an output pd,  $V_{out}$ , of 3.0V when unloaded (switch open). [2]

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- (b) Requirement **B**: a power dissipation of less than 0.50W in  $R_1$  when unloaded (switch open). [2]

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(c) Requirement **C**: a **decrease** of less than 0.40V in  $V_{\text{out}}$  when the switch is closed. [4]

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